

DATA CLEANING AND FEATURE ENGINEERING

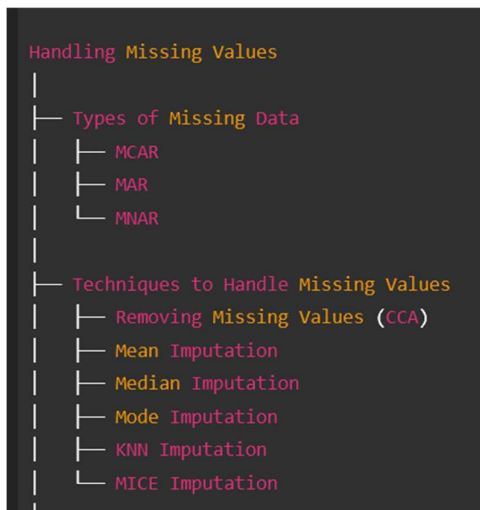
Data Pre-processing: It is the process of cleaning, transforming, and preparing raw data so that it can be effectively used by machine learning algorithms.

Why DPP is important → ML models learn from the data provided. Poor-quality data leads to poor-quality predictions (Garbage-in, Garbage-out).

Overview of DPP steps: Raw data → Remove duplicates → Handle missing values → Handle Outliers → Encode categorical variables → Scale numerical features → Feature Engineering → Model Ready Data

Handling Mixed Variables: A mixed variable is a column that contains more than one type of information in a single feature. Eg: cabin (Titanic) → C85 where Deck=C and Cabin Number = 85

REGEX OPERATORS: Regex (Regular expression) is a language used to find patterns inside index. **NOTE: REGEX is essential for FEATURE ENGINEERING**



CCA (Complete Case Analysis)/List wise deletion: Keep only the rows that have complete data. Delete row that contains a missing value. That's why it's called Complete Case.

If you want to modify the original DataFrame:

```
python
df.dropna(inplace=True)
```

When we use CCA:

- Only a few values are missing. Eg: 1000 rows and 20 rows have missing values. That is 5% of value only missing.
- Missing Values Are Random (MCAR) → Rows became miss randomly.

TYPES OF MISSING DATA:

- 1) **MCAR (Missing Completely At Random):** The missing value has nothing to do with any variables in the dataset. It disappeared completely by chance. MCAR is good because deleting those rows doesn't introduce bias.

Example 1: Hospital Records

A hospital stores patient information.

During a power outage, 20 patient records become corrupted. It completely happened by chance

- 2) **MAR (Missing At Random):** Here missing depends on another variable, but not on the missing value itself.

Dataset:

Gender	Salary
Male	50000
Female	NaN
Female	NaN
Male	60000

Women are less likely to disclose salary.

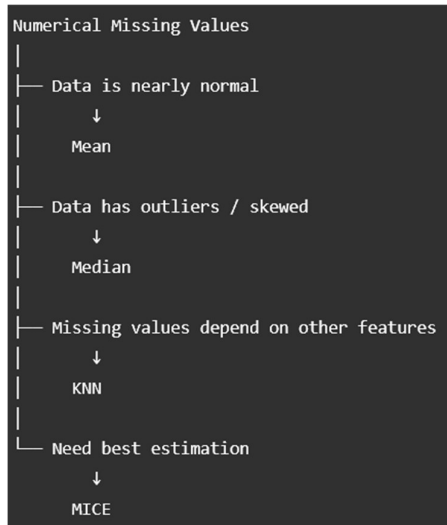
Missingness depends on GENDER.

- 3) **MNAR (Missing Not At Random):** Missingness depend on the value itself.

Handling Numerical Data:

- A) **Univariate Imputation:** Uses only one column to fill its own missing values.
 - Mean: Replace missing value with average.
 - Median: Replace missing value with middle vale.
 - Mode (Rarely used in Numerical data, since it is mostly used in categorical data): Replace missing value used most frequent value.
- B) **Multivariate Imputation:** Multiple columns are used together to fill missing values.
 - KNN Imputer (K Nearest Neighbors):

- MICE (Iterative Imputer) (Multiple Imputation by Chained Equation):



ARBITRARY VALUE IMPUTATION: Fill arbitrary values to fill missing values.

Common Arbitrary Values	
Feature	Arbitrary Value
Age	999 or -1
Salary	-999
Marks	-1
Temperature	-999
The chosen value should be outside the normal range.	

HANDLING CATEGORICAL MISSING DATA:

1) Univariate Imputation:

- Mode (Most Frequent) Imputation
- Missing Category Imputation: It is a technique where we replace missing values in a categorical column with a new category such as “MISSING” or “UNKNOWN” instead of using MODE
- Random Sample Imputation

2) Multivariate Imputation:

- KNN
- MICE (Less Common)

KNN imputer: (K Nearest Neighbors) imputation is a multivariate imputation technique that fills missing values by looking at the K most similar rows in the dataset and using their values to estimate the missing value.

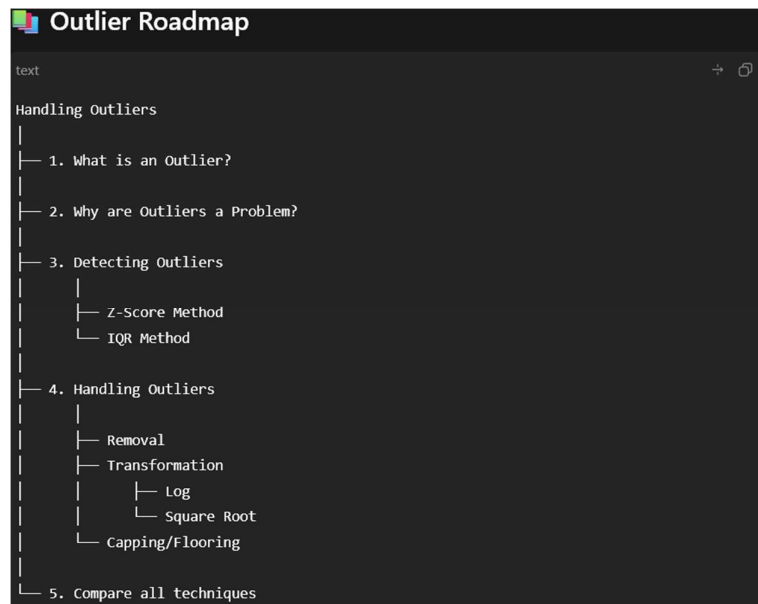
MICE IMPUTER/ ITERATIVE IMPUTER (Sci-kit learn): MICE is a multivariate imputation technique that predicts missing values by building a machine learning model for each feature with missing values, using the other features or predictors. This process is repeated multiple times until the estimate become stable.

Q: Why it is called CHAINED EQUATION?

ANS: Because each feature with missing values id predicted one after another in a chain, and the updated predictions are used to improve.

★ Easy trick to remember

- **Mean** → Uses **one value**.
- **KNN** → Uses **similar rows**.
- **MICE** → Uses **prediction models repeatedly**.

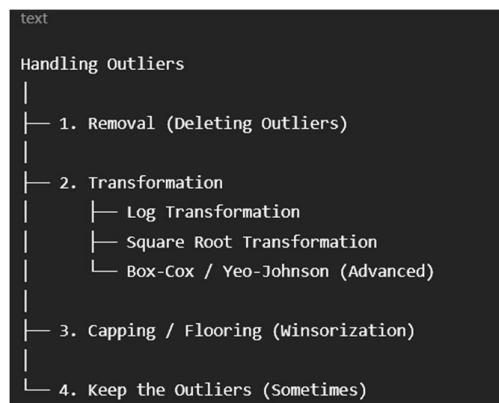


An **OUTLIER** is a data point that is significantly different from the rest of the observation in a dataset. Outliers can negatively affect machine learning models because they can change mean, affect model performance, produce incorrect predictions etc.,

Detecting Outliers: **NOTE→ Z-Score and IQR only work for numerical data**

- 1) **Z-score:** It is a statistical method used to detect outliers by measuring how many STD a data point is away from the mean. Z-SCORE tells us how far a value is from the mean in terms of standard deviation.
 - Use Z-SCORE when the data is approximately normally distributed (bell shaped).
 - If the data is heavily skewed, we can't use Z-Score.
- 2) **IQR (Inter Quartile Range):** A quartile divides the sorted data into 4 equal parts. IQR is the difference between Q3 and Q1. It represents the spread of the middle 50% of the data.
 - Q1→ First quartile – 25% of the data lies below Q1
 - Q2→ Second quartile – This is simply the median, 50% of the data lies below it.
 - Q3→ Third quartile – 75% of data lies below it
 - Uses when the data is highly skewed.

Classification of OUTLIER Treatment:



- A) **Removal:** Removing the rows containing outliers from the dataset. Eg: Trimming
- B) **Transformation:** It reduces the impact of outliers by changing the scale of the data instead of removing it. **NOTE: We don't remove the outlier – we shrink its effect.**
 - **Log Transformation→** A transformation technique that applies logarithm to numerical values to reduce the skewness and lessen the impact of large outliers.
 - **Square Root Transformation→** A transformation technique that applies square root to numerical values to moderately reduce the impact of outliers.
- C) **Capping / Flooring (Winsorization):** Replace extreme values with a maximum or minimum threshold instead of deleting them.

Outlier Treatment Methods

1. **Removal** → Delete outlier rows.
2. **Transformation** → Reduce the effect using Log or Square Root.
3. **Capping/Flooring** → Replace extreme values with thresholds.
4. **Keep** → Leave them unchanged if they are valid.

Capping: It is the process of replacing values that are above upper limit using upper limit itself

Flooring: It is the process of replacing values that are below lower limit using the lower limit itself.

CAPING & FLOORING are actually 2 parts of same method.

FEATURE SCALING: It is the process of bringing numerical features to similar range without changing their relative relationships.

Feature scaling changes the scale of numerical features while preserving their relative order and relationships.

Transformation

Transformation is the process of changing the mathematical distribution of a feature to reduce skewness, reduce the influence of outliers, or satisfy model assumptions.

Feature Scaling

Feature Scaling is the process of bringing numerical features to a common scale while preserving their relative order, making them suitable for machine learning algorithms.

SCALING TECHNIQUES:

- A) **Normalization (Min-Max Scaling)** → This transforms numerical values into a fixed range, usually between 0 and 1, while preserving their order.

Q: Why do some algorithms require feature scaling while others not?

Ans: Algorithms such as KNN, K-mean, SVM and neural networks depends on distances or optimization, so features with larger scales can dominate the learning process.

Feature scaling ensures all features contribute fairly.

Tree based algorithms like DECISION TREES and RANDOM FOREST split data based on feature threshold rather than distance, so they are generally unaffected by the scale of the features.

- B) **Standardization (Z-Score scaling)**→ It is the process of transforming numerical features so that they have a mean of 0 and a standard deviation of 1.

Algorithms like LOGISTIC REGRESSION, SVM, NEURAL NETWORKS train more efficiently because no feature dominates due to its scale.

NOTE: Standardization asks how many standard deviations away from mean is this value?

- C) **ROBUST SCALING:** Is a feature scaling technique that uses the median and IQR instead of mean and standard deviation, making it more robust to outliers. So, the extreme outliers have much less effect.

Interview Comparison		
Scaler	Uses	Affected by Outliers?
MinMaxScaler	Min & Max	✅ Yes (Highly)
StandardScaler	Mean & Standard Deviation	✅ Yes
RobustScaler	Median & IQR	❌ Much Less

NOTE:

Every scaler has 2 main goals. Let's compare them.			
Scaler	Goal 1	Goal 2	
MinMaxScaler (Normalization)	Make the minimum = 0	Make the maximum = 1	
StandardScaler	Make the mean = 0	Make the standard deviation = 1	
RobustScaler	Make the median = 0	Make the IQR = 1	

FEATURE ENGINEERING: It is the process of creating, transforming, or selecting features from raw data to improve the performance of machine learning models.

Q: Why feature engineering is important?

Ans: Better data beats a more complex algorithm.

Types of feature engineering:

- Creating new features
- Transformations
 - 1) Log Transformation
 - 2) Square root Transformations

- 3) **Polynomial Features**→ Polynomial feature engineering can create new features by combining existing features using power and interactions.

NOTE: The main idea of PF is “Can I create another feature that makes the relationship easier for a linear model to learn.”

- **Binning (Discretization):** It is the process of converting a continuous numerical feature into discrete categories (bins).
 - 1) Equal width binning: Each bin has equal width.
 - 2) Equal frequency binning: Each bin has roughly the same number of observations.
- **Encoding Categorical Variables:** It is the process of converting categorical data into numerical values so that ML algorithms can process it.
 - A) **Ordinal Categorical Data**→ CD that has a natural order or ranking between categories. Eg: Education level – High school<Graduate<Master<PhD.
 - 1) Ordinal Encoding
 - B) **Nominal Categorical Data**→ CD that has no natural order or ranking between categories. Eg: Green? Blue? Red?
 - 1) One Hot Encoding
 - C) **Binary Categorical Data**→ CD that has two categories only.
 - 1) Label Encoding: Is suitable when the categories can safely be represented by a single number

```
Encoding Categorical Variables
|
|— Ordinal Categorical Data
|   • Has natural order
|   • Example: Education Level
|   • Encoding: Ordinal Encoding
|
|— Nominal Categorical Data
|   • No natural order
|   • Example: Gender, Color, City
|   • Encoding: One-Hot Encoding
|
|— Binary Categorical Data
|   • Two categories only
|   • Example: Yes/No, Has Experience/No Experience
|   • Encoding: Label Encoding (or simple 0/1 mapping)
```

Q) Why we perform ENCODING after BINNING?

Ans: Binning converts numerical data into categorical groups. Since many ML algorithms cannot work directly with text categories, we encode these categories into numerical values before training models.

Q) Why is Label/Ordinal encoding suitable for ordinal data?

Ans: Because ordinal data has a natural ranking, assigning ordered numerical values preserves that ranking, allowing the model to use that order.

FEATURE SELECTION: It is the process of selecting most relevant features from a dataset while removing irrelevant, redundant, or noisy features to improve model performance.

Types of FEATURE SELECTION,

A) **Filter methods:** It select features based on statistical measures without training a ML model.

- 1) Correlation→ It measures the strength and direction of the relationship between two numerical variables.

NOTE: COVARIANCE tells you the direction of the relationship. Uses in PCA.

CORRELATION tells you both the direction and strength of the relationship. Uses in FEATURE SELECTION.

- 2) Variance threshold→ Removes features with low or zero variance, as they contain little or no useful information for ML model.

MULTINUCLEARITY: It occurs when two or more input features are highly correlated with each other, meaning they provide similar or redundant information's. So, we have to remove one of the columns

It is only about Feature ↔ Feature, if $\text{corr}=0.8, 0.9$ remove

B) **Wrapper method:** Are feature selection method that evaluate subsets of features by rapidly training a ML model and selecting the combination that gives the best performance.

- 1) Recursive Feature Elimination (RFE)→ That recursively removes the least important features based on a ML model until the desired number of features remains.
- 2) Stepwise selection→ **FORWARD SELECTION:** This starts with no features and adds one feature at a time based on which addition improve model performance most.

→ **BACKWARD ELIMINATION:** This starts with all features and removes the least useful feature one at a time based on model performance.

C) **Embedded methods:** Are feature selection technique that performs feature selection during the model training process by using models own feature importance or coefficients.

- 1) Lasso→ Regularization technique that shrink feature coefficients, with some coefficients become exactly zero, effectively removing those features from the model.
- 2) Ridge → Regularization technique that reduces the magnitude of feature coefficients to prevent overfitting while keeping all features in the model.

- Lasso = Shrink + Remove
- Ridge = Shrink + Keep

SUMMARY:



DATA LEAKAGE: It is the accidental use of information that the model should not know during training, resulting unrealistically high performance.

The Basic Idea

Imagine tomorrow you have an exam.

There are 100 questions.

Tonight, your teacher secretly gives you tomorrow's question paper.

Tomorrow you score

text

100/100

Question:

Are you really that good?

✗ No.

To avoid data leakage: Dataset → Train-Test Split → Everything Else.

NOTE: When to use Feature Scaling methods

StandardScaler

When to use:

- Data is approximately normally distributed.
- No major outlier problem (or outliers are not severe).
- Commonly used with Logistic Regression, SVM, KNN, Neural Networks.

MinMaxScaler

When to use:

- You want all values between 0 and 1.
- Common for Neural Networks and image data.

RobustScaler

When to use:

- There are **many influential outliers** that you plan to keep.
- It uses the **median and IQR**, making it less affected by extreme values.

NOTE: We are using heatmap to find whether 2 features carry same information. That is called MULTICOLLINEARITY.